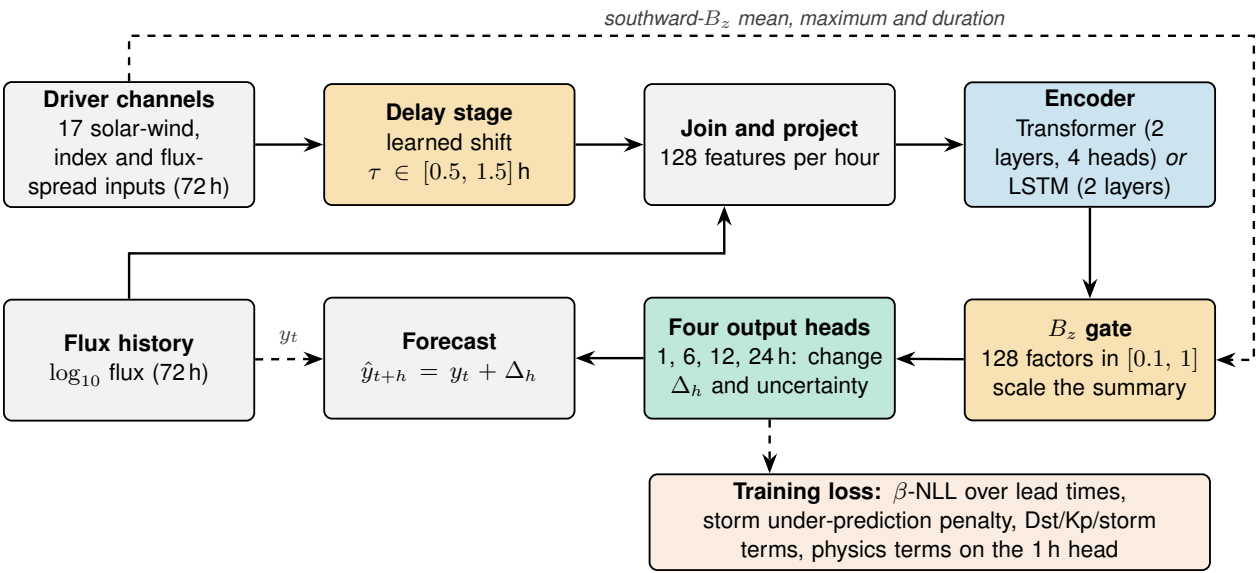

southward-B_z mean, maximum and duration
Driver channels
17 solar-wind, index and flux-spread inputs (72 h)
Delay stage
learned shift
$\tau \in [0.5, 1.5]$ h
Join and project
128 features per hour
Encoder
Transformer (2 layers, 4 heads) or LSTM (2 layers)
Flux history
$\log_{10}$ flux (72 h)
y_t
Forecast
$\hat{y}_{t+h} = y_t + \Delta_h$
Four output heads
1, 6, 12, 24 h: change Δ_h and uncertainty
B_z gate
128 factors in $[0.1, 1]$ scale the summary
Training loss: β-NLL over lead times, storm under-prediction penalty, Dst/Kp/storm terms, physics terms on the 1 h head